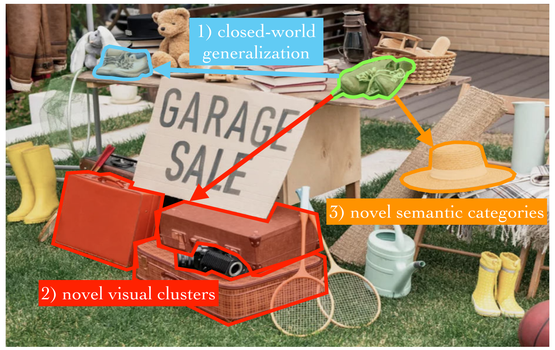
Open Ad-hoc Categorization with Contextualized Feature Learning

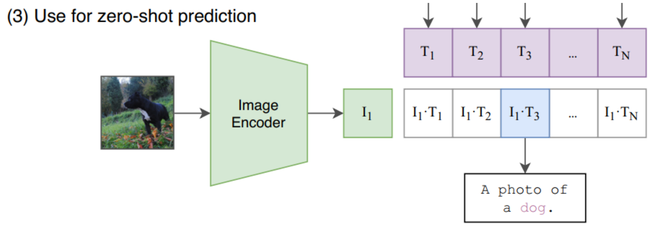
Zilin Wang * Sangwoo Mo * Stella X. Yu Sima Behpour Liu Ren

Is Categorization Solved?

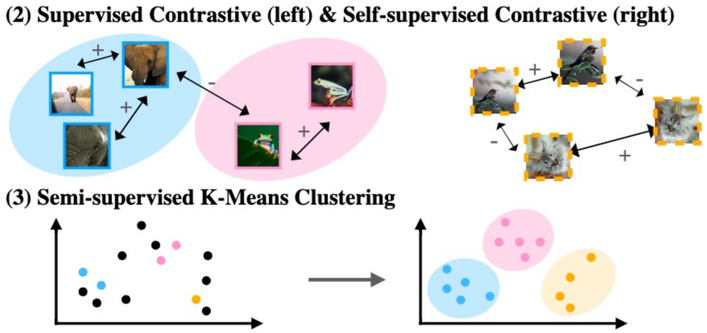


1. Ad-hoc categories (contexts)
2. Open-world

Baseline: CLIP w/ Extended Vocabulary



Baseline: Generalized Category Discovery

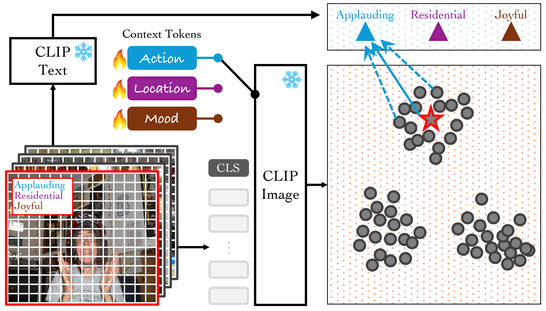


Open Ad-hoc Categorization (OAK)



Introduce Omni accuracy

Our OAK Framework

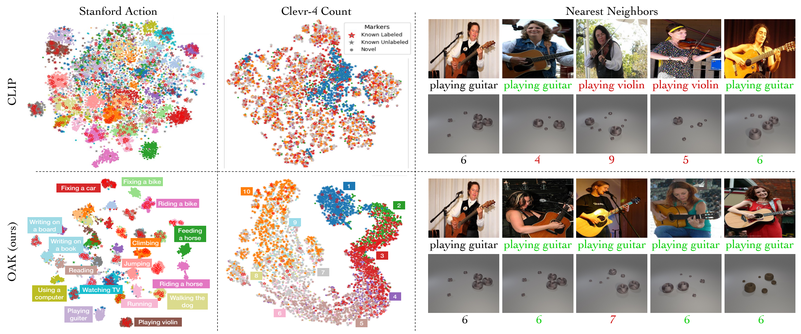


1. Based on CLIP
2. Tune context token
3. Top-down feedback from CLIP text encoder
4. Bottom-up clustering of visual features

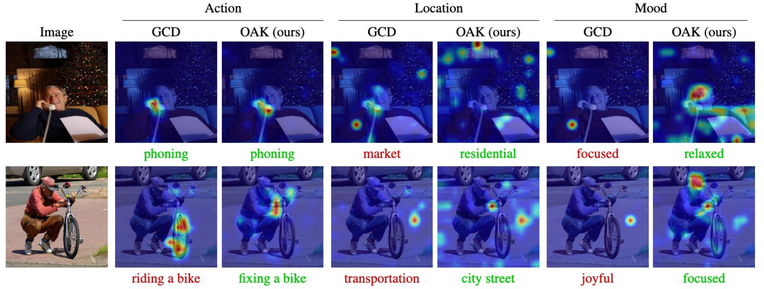
1. Novel Class Discovery & Omni Accuracy

Method	Known				Novel				Overall			
	Action	Location	Mood	Omni*	Action	Location	Mood	Omni	Action	Location	Mood	Omni
CLIP-ZS	**96.2**	60.4	**87.4**	75.0	-	-	-	-	-	-	-	-
+ LLM vocab	93.5	58.3	75.9	75.0	38.6	34.2	35.4	0.0	65.2	47.5	55.0	43.0
+ GT vocab	93.6	59.6	78.8	75.0	80.3	59.7	65.8	29.4	86.7	59.7	72.1	38.3
SS-KMeans	67.0	56.3	43.2	0.0	53.9	71.1	78.2	35.3	60.2	62.9	61.3	47.7
GCD	89.4	75.4	64.3	75.0	67.8	80.8	40.6	0.0	78.3	77.8	52.1	52.3
OAK (ours)	88.9	**83.9**	68.8	0.0	**85.1**	**88.4**	**87.4**	**47.1**	**86.9**	**85.9**	**78.4**	**70.3**

2. Contextualized Visual Embedding



3. Contextualized Visual Attention



4. Naming Novel Visual Clusters

a) Stanford	Action		Location		Mood	
GT Label	cooking	reading	restaurant or dining area	urban area or city street	focused	relaxed
Prediction	preparing a meal	reading a book	commercial kitchen	suburban street	explorative	admiring

b) Clevr-4	Texture		Color		Shape		Count	
GT Label	zigzag	circles	pink	cyan	star	diamond	4	9
Prediction	wavy lines	checkerboard	pink	turquoise	star shape	diamond shape	4	17

